



MS-AGDA: A multi-source adaptive gating unsupervised domain adaptation for cross-subject and cross-session EEG-based emotion recognition

Junqi Liu¹ · Yuliang Ma² · Michael Houston³ · Senda Gao¹ · Yingchun Zhang^{3,4,5}

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Abstract

Recognizing emotions is a vital component of applications related to Brain-Computer Interfaces (BCI). Although electroencephalography (EEG) offers rapid and reliable measurements, it is inherently limited by a low signal-to-noise ratio and non-stationarity, leading to variability across subjects and recording sessions. Existing multi-source domain adaptive strategies usually assume that the inter-domain distributions are static, ignoring the intrinsic variations in feature distributions between domains. In order to overcome the above limitations, this paper proposes an unsupervised multi-source domain adaptive network based on a gating mechanism, extracting unique features for each target domain. Furthermore, the model employs different loss functions for distributional alignment in the source and target domains to minimize the marginal and conditional distributions between the domains. Cross-subject experiments on the SEED, SEED-IV, and DEAP datasets were performed, while cross-session investigations were executed on the SEED and SEED-IV datasets. The proposed model improved classification performance, achieving cross-subject accuracy of 92.54%, 85.86%, and 65.59% on the SEED, SEED-IV, and DEAP datasets, respectively, and cross-session accuracy of 95.88% and 82.09% on the SEED and SEED-IV datasets.

Keywords Affective brain-computer interface · Domain adaptation · Emotion recognition · Transfer learning

1 Introduction

Emotion is a complex psychological state that has a profound impact on human behavior and decision-making processes [1]. Emotion recognition technology represents a core research topic in the field of affective computing, with applications in diverse fields including mental health monitoring, affective computing, and virtual reality. It is particularly noteworthy that affective Brain-Computer Interfaces (aBCI) have been developed, which bridge the gap between brain activity and computer systems for emotional interactions [2]. They show great potential in areas such as rehabilitation medicine and human-computer interaction. Despite this, the inherent complexity of EEG signals poses a considerable technical obstacle to effectively identifying and understanding human emotions. Accurately recognizing emotions generally requires examining both temporal and spectral characteristics obtained from EEG data. These fea-

✉ Yuliang Ma
mayuliang@hdu.edu.cn

✉ Yingchun Zhang
y.zhang@miami.edu

Junqi Liu
232320047@hdu.edu.cn

Michael Houston
m.jameshouston81@gmail.com

Senda Gao
yuiry159753@qq.com

¹ HDU-ITMO Joint Institute, Hangzhou Dianzi University, Hangzhou 310018, Zhejiang, China

² School of Automation, Hangzhou Dianzi University, Hangzhou 310018, Zhejiang, China

³ Department of Biomedical Engineering, University of Miami, Coral Gables 33146, Florida, USA

⁴ Desai Sethi Urology Institute, Miller School of Medicine, University of Miami, 1611 NW 12th Ave, Miami, FL 33136, USA

⁵ Miami Project to Cure Paralysis, Miller School of Medicine, University of Miami, 1611 NW 12th Ave, Miami, FL 33136, USA

tures can provide insights into the neural activity patterns of the brain in response to varying emotional states. It has been demonstrated that EEG signals evoked by different emotional states exhibit notable differences in the alpha, beta, and gamma wave bands [3]. Traditional EEG emotion recognition techniques often depend on manually derived statistical features and shallow machine learning algorithms, including techniques like Support Vector Machines (SVM) and Linear Discriminant Analysis (LDA) [4, 5].

In recent years, notable advancements in EEG emotion recognition have been achieved through deep learning approaches, with classical architectures like Convolutional Neural Networks (CNN) [6] and Long Short-Term Memory networks (LSTM) [7] being effectively employed to extract EEG signal features and conduct classification tasks. Given the dependence of deep learning-driven emotion recognition frameworks on extensive labeled EEG data, collecting and labeling EEG signals is time-consuming and labor-intensive, transfer learning and domain adaptation techniques are gradually becoming practical tools for solving cross-domain problems. [8, 9]. Domain adaptation techniques aim to improve model generalizability across domains by reducing distribution discrepancies between source and target feature spaces. Compared with Single-Source Domain Adaptation (SSDA) [10] that involves a single source-target domain pair, Multi-Source Domain Adaptation (MSDA) [11] has attracted growing attention due to its ability to leverage knowledge from multiple source domains. The fundamental concept behind MSDA is to develop a model that possesses enhanced generalization capabilities for the target domain by acquiring a shared representation from multiple source domains. However, many methods neglect to consider both marginal and conditional distribution differences in the domain feature alignment process, thus limiting the generalization ability of these models. Moreover, feature extraction methods for MSDA networks usually use fixed network parameters to couple domains with great individual differences, and are unable to dynamically adapt to the differences in feature distributions among different domains, and the dynamic adjustment mechanism is insufficient, leading to an over-reliance on hyper-parameter optimization.

To address the constraints of existing domain adaptation methods, we introduce a novel unsupervised multi-source domain adaptation framework for EEG emotion recognition, designated as the Multi-source Adaptive Gating Unsupervised Domain Adaptation Network (MS-AGDA). The model contains an adaptive gating mechanism that dynamically adjusts the intensity of feature extraction for a particular domain. In summary, the MS-AGDA framework integrates an innovative approach to unsupervised domain adaptation with adaptive feature extraction and a hierarchical domain alignment strategy. Fig. 1 summarizes the framework we propose.

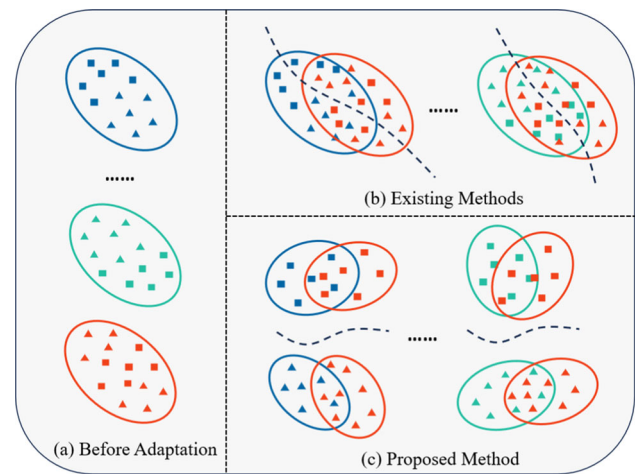


Fig. 1 Comparison between other domain adaptation methods and our method: **(a)** Domain shifts exists across multi-source and target domains. **(b)** Existing methods focus on directly aligning data from multi-source and target domains holistically, which may result in the misclassification of target samples due to a lack of dynamic feature extraction. **(c)** The proposed method employs an adaptive feature extraction process, thereby reducing the marginal and conditional distributional discrepancies between domains

The main contributions of this research can be outlined as follows:

- A novel multi-source unsupervised domain adaptive gating EEG emotion recognition network is proposed. The model employs Universal Domain Discrepancy (UDD), and Inter-Category Discrepancy (ICD), to narrow down the differences in marginal and conditional distributions between domains, respectively.
- An adaptive EEG signal feature extractor is constructed. The gating mechanism is employed to develop a method that can adaptively adjust the intensity of feature extraction for different domains, thereby enhancing the transferability of the model between source and target domains.
- Extensive experimental evaluations reveal that the proposed MS-AGDA achieves more advanced performance across SEED, SEED-IV, and DEAP datasets.

2 Related work

Recent EEG-based emotion recognition studies mainly adopt supervised deep learning, relying on labeled target-domain data. Bagherzadeh et al. [12] used Synchrosqueezing Wavelet Transform (SSWT) and ResNet-18 on two-channel EEG signals, achieving 75–81% accuracy across SEED datasets. Li et al. [13] fused single-channel Differential Entropy (DE) features and cross-channel Functional Connectivity (FC) features by Graph Neural Network (GNN) to extract higher-level graph structure information. Bagherzadeh et al. [14]

combined Continuous Wavelet Transform (CWT) with an ensemble of pre-trained CNNs and majority voting, achieving 96.9% accuracy on DEAP.

Nevertheless, while these supervised learning models achieve improved performance in laboratory settings, they remain limited in addressing distributional shifts across subjects and sessions. Domain adaptation (DA) techniques aim to bridge this gap by aligning the distributions between source and target domains. DA can be unsupervised, semi-supervised, or supervised, depending on whether target-domain labels are unavailable, partially available, or fully available. In EEG emotion recognition, target labels are often inaccessible, motivating unsupervised multi-source domain adaptation (UMSDA) approaches that leverage labeled source domains to adapt to unlabeled target domains [15]. Multi-Source Domain Adaptation (MSDA) offers an efficient method that leverages the abundant information from various source domains to enhance adaptation to the target domain. She et al. [11] designed an unsupervised emotion recognition framework based on a Multi-Source Associative Domain Adaptation (MSADA) network, incorporating both marginal and conditional domain distributions. Li et al. [16] introduced a Dynamic Domain Adaptation (DDA) framework, which gradually shifts focus from global to local subdomain variance and can be applied in both unsupervised and semi-supervised settings.

Although current BCI methods show potential in recognizing emotions from EEG data, they face considerable obstacles. The existing DA methods are deficient in terms of feature extraction and dynamic adjustment mechanisms when confronted with the intricacies of EEG signals (e.g.,

noise and nonlinear features), and are unduly reliant on hyperparameter settings. Moreover, many existing methods do not jointly address marginal and conditional distribution disparities during cross-domain feature alignment, constraining the model's generalization capacity. To address these issues, we propose MS-AGDA, an unsupervised multi-source domain adaptation framework inspired by Chen et al. [17]. MS-AGDA adaptively extracts domain-invariant and domain-specific features via UFD and SDFA modules, aligns multiple source domains sequentially with the target, and gradually reduces domain shifts. The integration of UDD and ICD losses further enhances generalization and mitigates both marginal and conditional distribution disparities in the unlabeled target domain.

3 Methods

To enhance the generalizability of EEG emotion recognition models in cross-subject and cross-session scenarios and mitigate domain bias caused by EEG signal distribution shifts across individuals and sessions, a framework comprised of three principal components was proposed: Universal Feature Downscaler (UFD), Specific Domain Feature Adapter (SDFA), and Specific Domain Classifiers (SDC). The framework diagram is presented in Fig. 2. The framework processes target domain and multiple source domain EEG inputs through the UFD to extract emotion-relevant universal features. Subsequently, these features undergo domain-specific adaptation via SDFA's dynamic gating mechanism, generating domain-specific emotional representations. The extracted

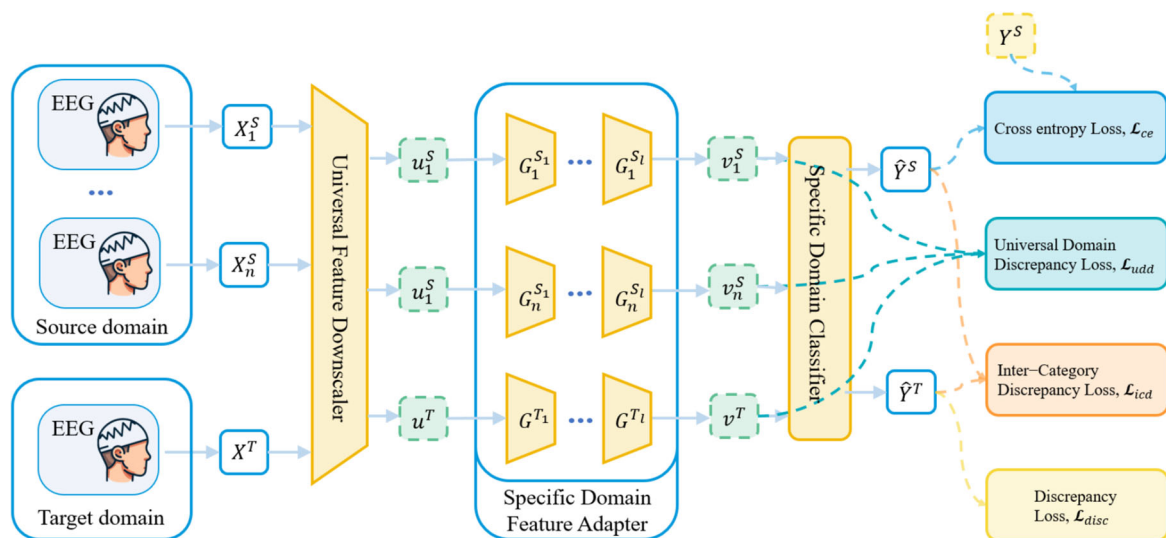


Fig. 2 An overview of the framework of our MS-AGDA network. In the figure, X stands for domain data, u stands for global invariant features, G stands for domain-specific feature extraction layer in the SDFA

module, v stands for domain-specific features, Y stands for data label, and $\hat{Y}$ stands for predicted data label

features are subsequently input into the SDC for emotion classification, with the resulting outputs applied to unsupervised learning in the source domain and performance assessment in the target domain.

3.1 Data preprocessing methodology

Differential Entropy (DE) serves as a widely utilized EEG feature extraction approach that captures the frequency properties of EEG signals across distinct frequency bands. Compared to the traditional spectral power features, the DE feature demonstrates high stability in emotion recognition tasks computed in the frequency domain and can maintain good feature consistency in cross-session and cross-subject tasks. Moreover, the distribution of the EEG signal within a fixed frequency band, which has been processed by appropriate band-pass filtering, can be approximated by a Gaussian distribution [18]. Therefore, it is assumed that the EEG signal obeys a Gaussian distribution $N(\mu, \sigma^2)$ within each frequency band. Under this assumption, the DE feature can be expressed as:

$$\begin{aligned} DE &= - \int f(x) \log(f(x)) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \log \left[\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right] dx \\ &= \frac{1}{2} \log 2\pi e \sigma^2 \end{aligned} \quad (1)$$

For both SEED and SEED-IV datasets, DE features are extracted across five EEG spectral bands: delta (δ : 1-3 Hz), theta (θ : 4-7 Hz), alpha (α : 8-13 Hz), beta (β : 14-30 Hz), and gamma (γ : 31-50 Hz).

3.2 Universal feature downscaler

The Universal Feature Downscaler (UFD) module is employed to extract global features of the EEG signal with the objective of capturing domain-invariant base information. This approach serves to enhance the model's generalization capability in cross-domain tasks. The UFD module comprises two principal components: a feature downscaling module and a Fast Fourier Transform (FFT) module. The feature downscaling module consists of a linear layer with 310 input neurons in the SEED and SEED-IV datasets, 160 input neurons in the DEAP dataset, and 64 output neurons in both, a batch normalization layer, followed by a LeakyReLU activation function. The features are divided into two sets for the purpose of reducing their dimensionality. One of these sets is processed by the FFT, and the two sets are then connected by means of a concatenation method. The input is a DE feature matrix, and the output after UFD is a generalized

feature representation after dimensionality reduction. This ensures the module can efficiently extract and retain consistent feature information across domains. The features output from the UFD will be used as inputs for subsequent modules.

3.3 Specific domain feature adapter

The Specific Domain Feature Adapter (SDFA) is a domain-specific feature adaptation module in the model, aiming at adaptive feature conditioning of the input features according to the properties of different source and target domains. Inspired by the Gated Recurrent Unit (GRU) network [19], SDFA consists of a series of dynamic gating layers and nonlinear activation functions. By dynamically adjusting the contributions of each layer, SDFA can automatically adjust the feature extraction strength of each layer according to the different domain characteristics of the input features, thus enhancing the representation of domain-specific features. The following pseudocode in Algorithm 1 provides a detailed overview of the SDFA module's operations. In each layer, a gating parameter g_i is computed to determine the extent of the layer's contribution to the final feature representation, balancing between the current layer's output and the previous hidden state.

Algorithm 1: Specific Domain Feature Adapter (SDFA)

Input: Input feature vector x , number of layers l , initial weights and biases W_0, b_0 , weights and biases for each layer W_i, b_i , and gating parameters W_g, b_g .

Output: Adaptively updated feature representation h_l .

```

1 Initialization: Compute initial feature transformation with ReLU
  activation:  $h \leftarrow \text{ReLU}(W_0 \cdot x + b_0)$ 
2 for  $i = 1$  to  $l$  do
3   Compute layer output:  $h_i \leftarrow \text{ReLU}(W_i \cdot h_{i-1} + b_i)$ 
4   Calculate gating parameter  $g_i$ :  $g_i \leftarrow \sigma(W_g \cdot h_i + b_g)$ 
5   Update hidden state using gating mechanism:
      $h_i \leftarrow g_i * h_i + (1 - g_i) * h_{i-1}$ 
6   Note: If  $g_i \approx 1$ , layer  $i$  contributes more strongly; if  $g_i \approx 0$ ,
     it relies more on  $h_{i-1}$ .
7 end
8 Return:  $h_l$ , the final adaptive feature representation for
  domain-specific input features.
```

In this algorithm, the gating parameter g_i is controlled by the sigmoid function, represented by the variable σ , which limits the output to the interval $[0, 1]$. This determines the degree of mixing between the output of each layer and the hidden state of the previous layer. This dynamic gating mechanism enables the model to adapt the depth of feature extraction in accordance with the specific requirements of a given domain, thereby facilitating the generation of an optimized, domain-specific representation, h_l , for each input domain. Dynamic changes in gating parameters allow the model to adaptively select the output of the current layer or

retain the state of the previous layer and subsequently update the state of each layer.

The purpose of computing the loss function UDD is to match the marginal distributions of the source and target domains by reducing the latent space between them, thus ensuring that the feature representations of different domains are in the same feature space. This alignment is quantified via MMD [20], which measures distribution divergence by comparing mean embeddings in a Reproducing Kernel Hilbert Space (RKHS). For layer l , the UDD loss is computed as specified in (2). During training, optimization of this loss progressively decreases the inter-domain distance in the feature space.

$$\begin{aligned} \mathcal{L}_{udd} = & \frac{1}{n_s^2} \sum_{i=1}^{n_s} \sum_{i=1}^{n_s} k_l(\phi_l(x_i^s), \phi_l(x_j^s)) \\ & + \frac{1}{n_t^2} \sum_{i=1}^{n_t} \sum_{i=1}^{n_t} k_l(\phi_l(x_i^t), \phi_l(x_j^t)) \\ & - \frac{2}{n_s n_t} \sum_{i=1}^{n_s} \sum_{i=1}^{n_t} k_l(\phi_l(x_i^s), \phi_l(x_j^t)), \end{aligned} \quad (2)$$

where x_i^s and x_j^t denote the i -th source domain sample and j -th target domain instance, respectively, n_s and n_t represent the number of samples in the source and target domains, and k_l represents the chosen kernel function of the l th layer.

Inspired by the contrastive domain discrepancy [20], we propose ICD to reduce the difference in conditional distributions between the source and target domains to ensure more consistent outputs for the same class of samples under different domains. For any two categories a_1 and a_2 , the distance $\hat{G}_{a_1 a_2}(\hat{y}_{1:n_t}^t; f)$ between the source and target domains is calculated as follows:

$$\hat{G}_{a_1 a_2}(\hat{y}_{1:n_t}^t; f) = g_1 + g_2 - 2g_3 \quad (3)$$

Among them, g_1 represents the intra-class distance of category a_1 in the source domain, g_2 denotes the intra-class distance of category a_2 in the target domain, and g_3 quantifies the cross-domain distance between category a_1 in the source domain and category a_2 in the target domain. The specific g_1 , g_2 , and g_3 are defined as follows:

$$g_1 = \frac{\sum_{i=1}^{n_s} \sum_{j=1}^{n_s} m_{a_1 a_1}(y_i^s, y_j^s) k(f(x_i^s), f(x_j^s))}{\sum_{i=1}^{n_s} \sum_{j=1}^{n_s} m_{a_1 a_1}(y_i^s, y_j^s)} \quad (4)$$

$$g_2 = \frac{\sum_{i=1}^{n_t} \sum_{j=1}^{n_t} m_{a_2 a_2}(\hat{y}_i^t, \hat{y}_j^t) k(f(x_i^t), f(x_j^t))}{\sum_{i=1}^{n_t} \sum_{j=1}^{n_t} m_{a_2 a_2}(\hat{y}_i^t, \hat{y}_j^t)} \quad (5)$$

$$g_3 = \frac{\sum_{i=1}^{n_s} \sum_{j=1}^{n_t} m_{a_1 a_2}(y_i^s, \hat{y}_j^t) k(f(x_i^s), f(x_j^t))}{\sum_{i=1}^{n_s} \sum_{j=1}^{n_t} m_{a_1 a_2}(y_i^s, \hat{y}_j^t)} \quad (6)$$

$$m_{a_1 a_2}(y_i, \hat{y}_j) = \begin{cases} 1, & \text{if } y_i = a_1 \text{ and } \hat{y}_j = a_2 \\ 0, & \text{else} \end{cases} \quad (7)$$

The kernel function $k(f(x_i), f(x_j))$ serves to compute the similarity of samples x_i and x_j in the feature space and is used to measure the distance between pairs of samples.

In this work, we employ an iterative clustering algorithm based on feature centers to generate pseudo-labels for the target domain. First, specified feature vectors are extracted from the network and cluster centers are initialized. Subsequently, the target domain samples are assigned to the closest cluster centers by batches and the center positions are updated by weighted average while keeping the initialized centers to prevent empty clusters. The process is iterated until the centers converge or change below a set threshold. To ensure the consistency of clusters and categories, we use the Hungarian algorithm to align the cluster centers. Eventually, each sample is assigned to the optimal cluster and saved as a pseudo-label for subsequent conditional distribution alignment optimization for ICD loss. The method eliminates the need for target domain labels, achieves completely unsupervised cross-domain pseudo-label generation, and effectively enhances the generalization ability of the model.

In summary, the ICD loss is used to align the conditional distributions of different classes in the source and target domains to ensure consistency of the same class of features across domains. Inspired by the contrastive domain discrepancy, the ICD loss is computed as follows:

$$\begin{aligned} \mathcal{L}_{icd} = & \frac{1}{M} \sum_{c=1}^M \hat{g}_{cc}(\hat{y}_{1:n_t}^t; f) \\ & - \frac{1}{M(M-1)} \sum_{c=1}^M \sum_{c' \neq c}^M \hat{g}_{cc'}(\hat{y}_{1:n_t}^t; f) \end{aligned} \quad (8)$$

The term " $\hat{g}_{cc}$ " is used to quantify the discrepancy in the distribution of features for category c in the source and target domains. This term ensures that samples belonging to the same category are characterized in a consistent manner between the two domains. $\hat{g}_{cc'}$ measures the difference in the distribution of features across categories c and c' in the source and target domains. This term ensures that the feature distributions of different categories remain separate in the source and target domains to enhance the discrimination between categories. The term M denotes the number of classes. The pseudo label of the samples in the target domain, denoted by $\hat{y}_{1:n_t}^t$, is obtained through the application of the feature extraction function, f , of the deep neural network.

3.4 Specific domain classifier

The Specific Domain Classifier (SDC) is the domain-specific classification module of the model, which is used to classify emotions based on specific domain features. Each one-to-one source and target domain branch has an independent softmax classifier, and these classifiers make classification predictions based on the features extracted from the SDFA separately. During the training phase, the model output distribution is closely approximated to the true distribution through cross-entropy loss (9), thereby directly optimizing classification accuracy. Since the classifiers are trained independently on their respective source domains, the data distributions of different source domains can lead to potentially inconsistent prediction results of each classifier on the target domain. Especially in samples close to the decision boundary, multiple classifiers may give different class predictions, leading to a large uncertainty in the average result near the boundary. Therefore, discrepancy loss (10) is introduced to reduce the classification uncertainty by reducing the output inconsistency between classifiers, making the final integrated prediction more robust. The mean of the predictions made by the N classifiers is taken as the final result.

$$\mathcal{L}_{ce} = \sum_{i=1}^N E_{x \sim X_S} J(\hat{y}_i^S, y_i^S) \quad (9)$$

$$\mathcal{L}_{disc} = \sum_{i \neq j}^N E_{x \sim X_i} |\hat{y}_i^T - \hat{y}_j^T| \quad (10)$$

In this context, the symbols y_i^S and y_j^S represent the true labels of the i th and j th source domain samples, respectively. These are known during the training phase. The model's predicted labels for these source domain samples are represented by $\hat{y}_i^S$ and $\hat{y}_j^S$, respectively. For the target domain, $\hat{y}_i^T$ and $\hat{y}_j^T$ represent the model's predicted labels for the i th and j th target domain samples.

The final loss function is a combination of cross-entropy loss, discrepancy loss, UDD loss, and ICD loss, as represented by the following formula:

$$\mathcal{L} = \mathcal{L}_{ce} + \alpha \mathcal{L}_{disc} + \beta \mathcal{L}_{udd} + (1 - \beta) \mathcal{L}_{icd} \quad (11)$$

The overall objective of the loss function is to promote feature alignment across source and target domains. During experiments, the balancing coefficients α and β are defined as $\alpha = \frac{1 - e^{-10 \cdot i / epoch}}{1 + e^{-10 \cdot i / epoch}}$ and $\beta = \frac{\alpha}{100}$, respectively, which regulate the contribution of each loss component.

4 Experiments and results

4.1 Datasets

1. SEED Dataset [3]: An important dataset in EEG studies, comprising EEG recordings from 15 subjects. Fifteen 4-minute Chinese film clips were selected and designed to induce three emotional states: positive, neutral, and negative.
2. SEED-IV Dataset [21]: The SEED-IV dataset is an extended version of the SEED dataset with a more nuanced division of emotion categories. The dataset comprised 15 subjects, with the emotional states divided into four categories: happy, sad, fear, and neutral.
3. DEAP Dataset [22]: The DEAP dataset contains EEG and peripheral physiological signals from 32 subjects. In contrast to the previous datasets, DEAP utilized a continuous emotion rating system across four dimensions: arousal, valence, dominance, and liking, each rated on a 1-9 scale. In this study, we used the widely used binary classification processing scheme: the binary classification problem was constructed by labeling the valence scores greater than 5 as high valence (labeled as 1) and those less than or equal to 5 as low valence (labeled as 0).

4.2 Experiments settings

This section details each model module's specific parameter settings and implementation details. Pre-processed DE features were utilized instead of raw data for signaling in the SEED and SEED-IV datasets. The UFD module consists of two parallel paths, which first reduce the dimensionality from 310 to 64 utilizing a linear layer, then employ a batch normalization layer and a LeakyReLU activation function. After one of the paths extracts the frequency domain features through the FFT module, the two paths are merged through the concatenation method, and the dimensionality is raised to 128. Then, the dimensionality is reduced from 128 to 64 through a fully connected layer. The SDFA module is structured as an initial layer followed by l subsequent layers, with the initial layer converting the input dimensionality from 64 to 32 and each subsequent layer retaining 32 dimensions.

The module's core lies in its dynamic gating mechanism. Each layer is equipped with trainable gating parameters, and a Sigmoid function maps the gating values between 0 and 1 in the forward propagation to dynamically control the level of participation in the current layer's output. In each layer, the current layer output is processed by the activation function ReLU and combined with the processed output of the previous layer, the weighting of which is determined by the gating parameter. A gating value approaching 1 indicates that the output of the current layer is given a greater weight.

Table 1 Cross-subject Emotion Recognition Results On SEED and SEED-IV Dataset

Dataset	Method	Acc(%)	F1-score	Kappa	Time(s)
SEED	DDC	57.45 ± 8.03	0.49	0.36	4210.32
	DCORAL	57.62 ± 9.81	0.53	0.38	982.47
	DAN	56.67 ± 8.07	0.46	0.33	1186.05
	MS-MDA	89.63 ± 6.79	0.83	0.75	9120.84
	SDDA	91.08 ± 7.70	—	—	—
	MS-AGDA(Ours)	92.54 ± 4.10	0.87	0.81	10125.47
SEED-IV	DDC	70.13 ± 11.00	0.69	0.59	9215.27
	DCORAL	37.43 ± 3.08	0.35	0.23	1695.44
	DAN	70.27 ± 10.42	0.69	0.60	1852.60
	MS-MDA	59.34 ± 5.48	0.57	0.48	10421.77
	SDDA	81.58 ± 8.72	—	—	—
	MS-AGDA(Ours)	85.86 ± 2.60	0.67	0.57	12840.56

Table 2 Cross-session Emotion Recognition Results On SEED and SEED-IV Dataset

Dataset	Method	Acc(%)	F1-score	Kappa	Time(s)
SEED	DDC	68.24 ± 10.37	0.63	0.52	4982.10
	DCORAL	65.61 ± 11.63	0.62	0.50	1015.28
	DAN	67.19 ± 10.42	0.61	0.50	1253.66
	MS-MDA	88.56 ± 7.80	0.80	0.69	9342.11
	SDDA	92.89 ± 7.64	—	—	—
	MS-AGDA(Ours)	95.88 ± 4.24	0.87	0.81	11234.65
SEED-IV	DDC	61.21 ± 11.07	0.56	0.47	9231.52
	DCORAL	44.63 ± 11.38	0.42	0.31	1690.25
	DAN	61.04 ± 12.33	0.57	0.47	1842.13
	MS-MDA	61.43 ± 15.71	0.59	0.48	11220.65
	SDDA	80.82 ± 10.43	—	—	—
	MS-AGDA(Ours)	82.09 ± 11.88	0.67	0.57	13820.44

In contrast, a gating value approaching 0 indicates that the previous layer's output is given a smaller weight. The parameters are continuously updated during training to achieve feature extraction adaptivity. The SDFa module generates a 32-dimensional output feature vector containing multi-level information extracted from each layer through a dynamic layer-by-layer weighting process. Finally, the SDC consists of a linear layer with dimensions reduced from 32 to the classification dimension.

4.3 Main results

In both cross-subject and cross-session experiments, Leave-One-Out Cross-Validation (LOOCV) is employed. This strategy employs the remaining subject data for model training, with one subject-by-subject training data set serving as the test set. During training, the selected subject's data served as the test set while the remaining subjects comprised the training set. This procedure was repeated iteratively to ensure reliability. Here are the models used as a comparison: DDC

Table 3 Cross-subject Emotion Recognition Results On DEAP Dataset

Method	Acc(%)	F1-score	Time(s)
DDC	58.76 ± 6.19	0.42	1284.13
DCORAL	59.43 ± 5.82	0.43	389.72
DAN	59.25 ± 5.98	0.44	403.74
TCN + ADDA	64.33 ± 7.06	—	—
MS-MDA	59.68 ± 7.11	0.48	10902.28
MS-AGDA(Ours)	65.59 ± 5.50	0.45	13562.30

[23], DCORAL [24], DAN [25], TCN + ADDA [26], MS-MDA [17], and SDDA [16]. As shown in Tables 1, 2 and 3.

Given that the SEED and SEED-IV datasets comprise 15 subjects each, we randomly select 15 subjects from the DEAP dataset. As illustrated in Table 3, the MS-AGDA model achieved an accuracy of 65.59% ± 5.50% on the DEAP dataset.

Table 4 Ablation Results(%) On SEED, SEED-IV and DEAP Dataset

Dataset	Method	Cross-subject	Cross-session
SEED	Our full	92.54 ± 4.10	95.88 ± 4.24
	w/o. Disc loss	86.25 ± 8.40	94.39 ± 5.50
	w/o. UDD loss	83.17 ± 6.14	85.78 ± 6.13
	w/o. ICD loss	85.74 ± 8.78	94.00 ± 5.77
	w/o. UFD	84.71 ± 7.79	92.72 ± 5.70
	w/o. SDFA	84.91 ± 6.42	89.75 ± 6.69
	Baseline	81.49 ± 7.37	82.32 ± 7.69
SEED-IV	Our full	85.86 ± 2.60	82.09 ± 11.88
	w/o. Disc loss	69.03 ± 10.40	65.72 ± 12.85
	w/o. UDD loss	65.05 ± 7.33	56.85 ± 8.57
	w/o. ICD loss	65.63 ± 9.37	64.92 ± 12.19
	w/o. UFD	46.83 ± 5.87	56.98 ± 9.40
	w/o. SDFA	58.52 ± 12.68	61.41 ± 14.22
	Baseline	36.81 ± 8.16	45.15 ± 12.15
DEAP	Our full	65.59 ± 5.50	—
	w/o. Disc loss	58.65 ± 7.73	—
	w/o. UDD loss	59.61 ± 8.78	—
	w/o. ICD loss	59.31 ± 8.67	—
	w/o. UFD	59.12 ± 7.74	—
	w/o. SDFA	59.92 ± 7.07	—
	Baseline	56.20 ± 7.62	—

The experimental results show that the MS-AGDA model performs well on several assessment metrics. In terms of F1 scores and Kappa coefficients, MS-AGDA has some advantages over other methods, showing relatively better classification consistency and reliability. In terms of computational efficiency, DCORAL and DAN have shorter training time, which is suitable for scenarios requiring higher computational efficiency. MS-AGDA has relatively longer training time, which is the computational price to pay for better performance. The stable performance of MS-AGDA under different experimental conditions proves the reasonableness of its training time.

The ablation experiments analyze the contribution of each loss function and module to model performance by progressively removing them and observing the effect on model performance on the SEED, SEED-IV, and DEAP datasets, as shown in Table 4.

4.4 Visualization

Figure 3 illustrates the trend of loss change with iterations under the cross-subject setting. The first subject served as the test set in the experiment, with the remaining 14 subjects constituting the training source domain. The graph shows the loss value variation for the second source domain. The loss variation plots include (a) Cross Entropy Loss, (b) Discrep-

ancy Loss, (c) UDD Loss, (d) ICD Loss. As the number of iterations reaches 500, the values of all the loss functions converge and decrease, reflecting that the training process of the model tends to stabilize.

The confusion matrices on the SEED and SEED-IV datasets are shown in Fig. 4 to visualize the difference in classification performance. The first row, Fig. 4a, b and c, shows the confusion matrices of the DDC algorithm, the MS-MDA model (baseline), and our model on the SEED dataset in turn; the second row, Fig. 4d, e and f, shows the corresponding results of the three models on the SEED-IV dataset.

5 Conclusion

In this study, we proposed MS-AGDA, a multi-source domain adaptation framework for EEG-based emotion recognition which achieves advanced performance on SEED, SEED-IV, and DEAP datasets. The gated network of SDFA can adjust the intensity of feature extraction using domain-specific input data, thereby enabling the adaptive adjustment of feature extraction weights at each layer of the network to generate domain-specific feature representations in multi-source domain adaptation tasks. Combining UDD and ICD losses enables the model to reduce both marginal and conditional distribution discrepancies simultaneously, thereby enhancing cross-domain generalization. Furthermore, the gradual convergence of losses during training is demonstrated through loss change curves.

The MS-AGDA framework can be applied to the field of emotional state monitoring in clinical aided diagnosis, as well as emotion perception in next-generation human-computer interaction. As it is based on the transfer learning paradigm, the model avoids the direct reliance on raw EEG data to a certain extent, which has potential privacy-preserving advantages. However, ethical concerns are still needed for practical deployment, including the security and anonymization of EEG data, the interpretability of model decisions, and the prevention of misclassification due to algorithmic bias.

However, this study still has some limitations. First, the current framework mainly focuses on DE features and lacks explicit modeling of temporal dynamics and spatial topology. Hu et al. [27] proposed a hybrid spatiotemporal feature, an emotion recognition network, that considers the complementarity between the two features. Ignoring these multidimensional features may limit the model's discriminative ability in complex scenes. Second, MS-AGDA is higher in terms of model complexity and computational cost, as it needs to process multiple source domains in parallel and maintain multiple sub-modules, resulting in higher training and inference resource consumption. In dataset like DEAP, MS-AGDA only shows limited performance enhancement, which is not a clear advantage over traditional non-adaptive

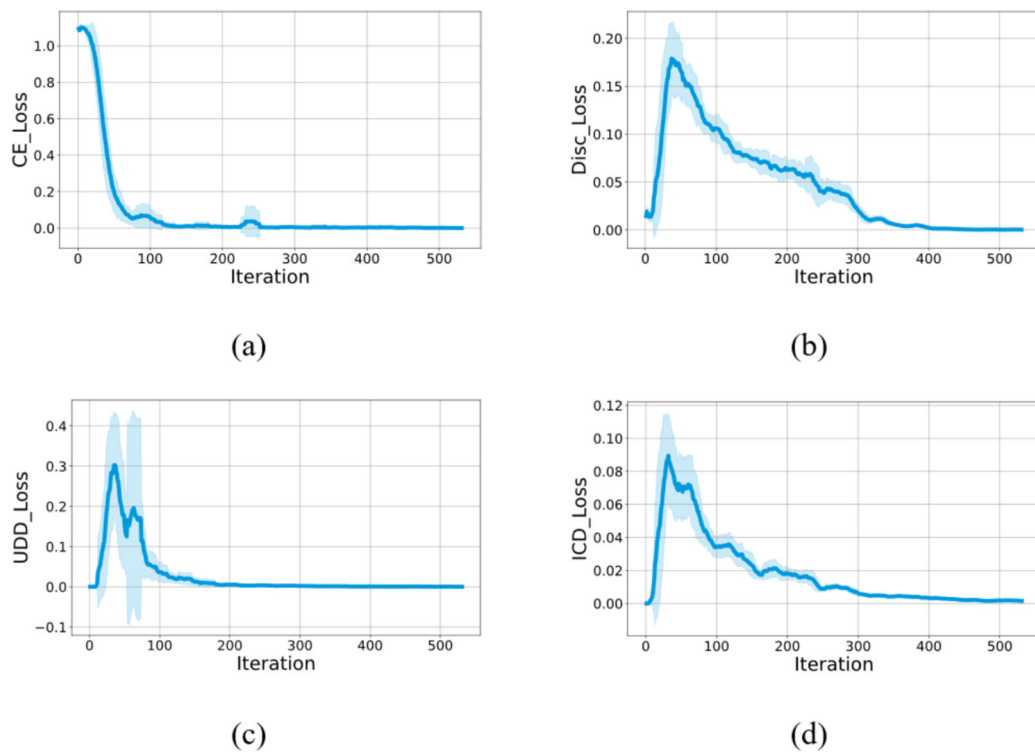


Fig. 3 Loss images for each iteration. (a) represents the cross-entropy loss, (b) the discrepancy loss, (c) the UDD loss, (d) the ICD loss

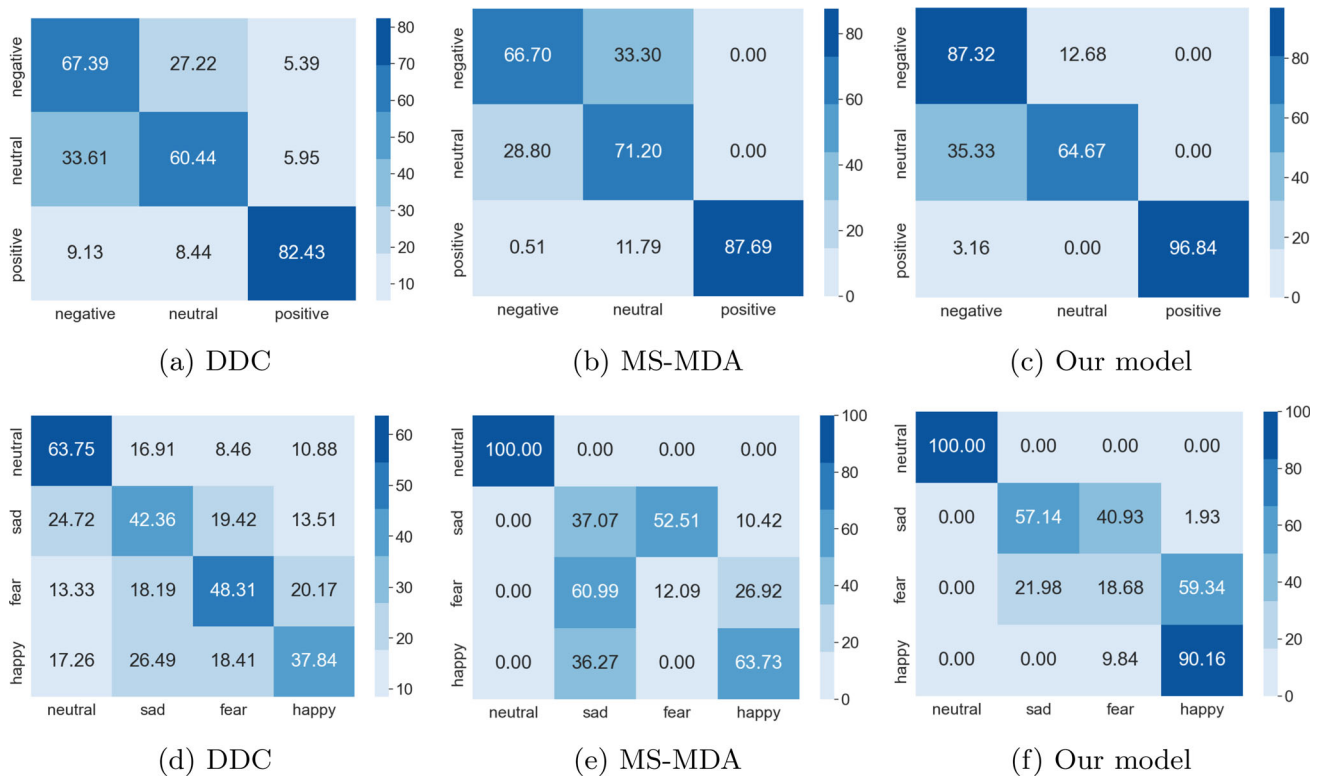


Fig. 4 Confusion matrices for the SEED and SEED-IV datasets

methods, and this also affects its practical applicability to a certain extent. In addition, the method proposed in this paper still relies on source domain labeling, while the attention-based passive multi-domain adaptive framework proposed by Hesham et al. [28] achieves adaptation without source labeling.

In future work, we will aim to introduce spatio-temporal feature extraction mechanisms, such as embedding temporal convolution or spatial attention modules, to more comprehensively characterize the dynamic and structural information of EEG signals. We will also optimize the model structure and simplify the computational process to reduce the computational overhead and validate its generalization ability on richer datasets. In addition, we plan to conduct cross-dataset experiments, such as migration experiments from SEED to DEAP, to more rigorously validate the robustness and scalability of MS-AGDA under different experimental paradigms and different acquisition conditions.

Author Contributions Liu and Ma performed model formulation, experimental validation, and first draft writing. Michael and Gao provided critical advice and support during the paper writing process. Zhang provided research direction and project management.

Data Availability The data generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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